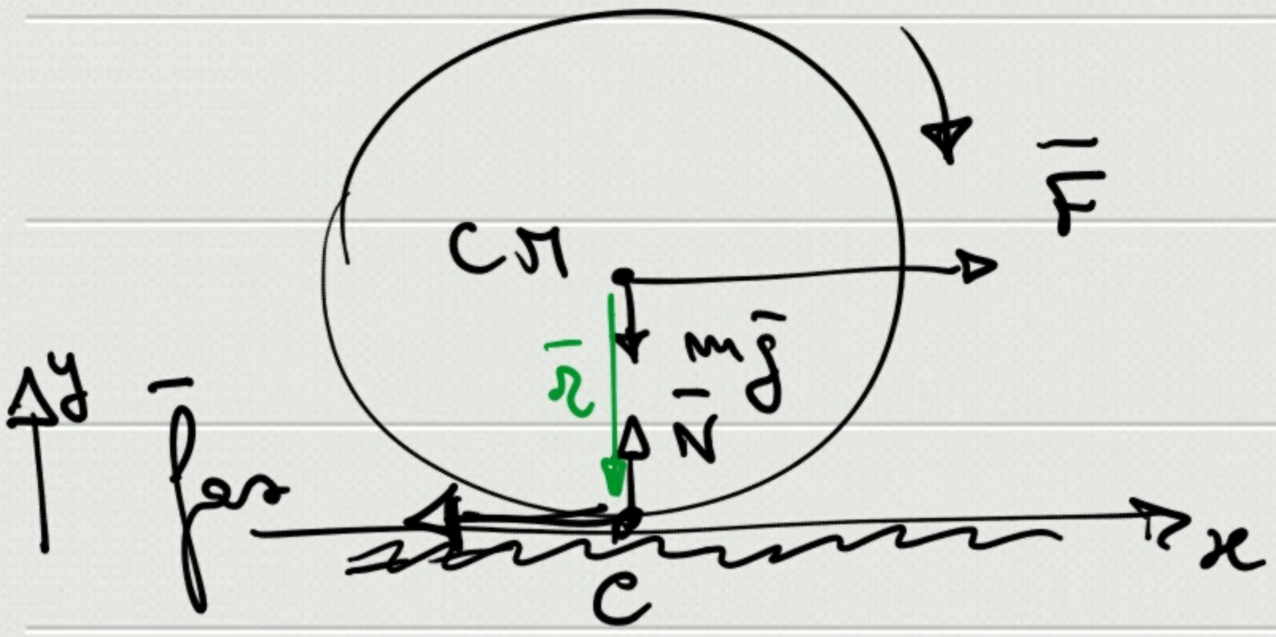


$$\bar{R}^{E} = m\,\bar{a}_{CM}$$

$$\bar{M}^{E}_{polo} = \bar{I}_{asse}\,\bar{\alpha}$$

$$\bar{M}_{O} = \bar{r} \times \bar{F}$$

$$\bar{F} + m\bar{g} + \bar{N} + \bar{f}_{ar} = m\,\bar{a}_{CM}$$

$$\text{polo} \equiv CM \Rightarrow \underset{\otimes}{\bar{R} \times \bar{f}_{ar}} = \bar{I}_{CM}\,\bar{\alpha}$$

$$\Rightarrow \begin{cases} F - f_{ar} = m\,a_{CM} \\ N - mg = 0 \qquad \Rightarrow N = mg \\ R\,f_{ar} = I_{CM}\,\alpha \\ a_{CM} = \alpha R \end{cases}$$

$$\Rightarrow R\,f_{ar} = I_{CM}\,\alpha = I_{CM}\,\frac{a_{CM}}{R} \Rightarrow f_{ar} = \frac{I_{CM}}{R^2}\,a_{CM}$$

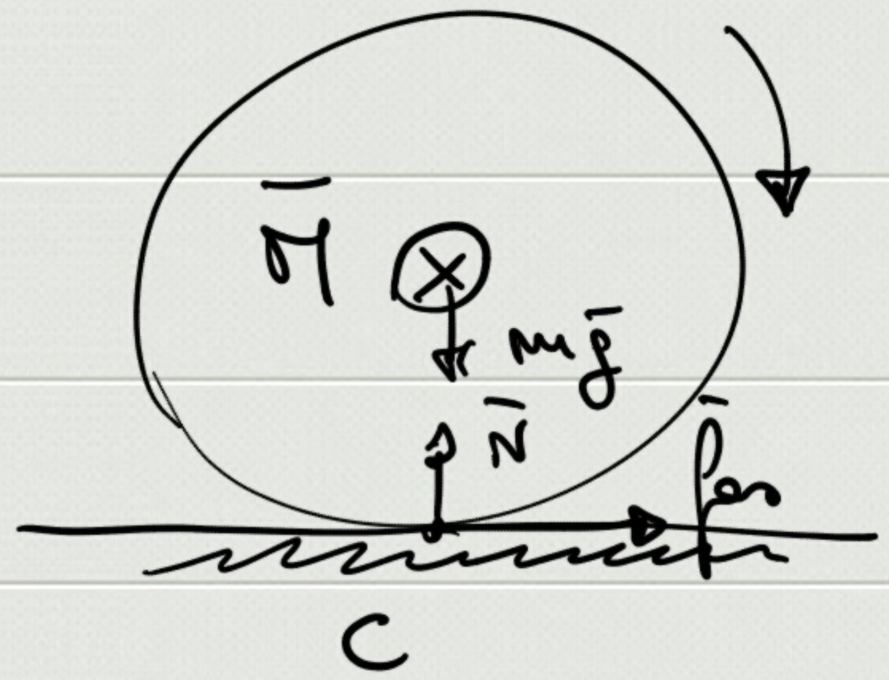
$$\Rightarrow F - \frac{I_{CM}}{R^2}\,a_{CM} = m\,a_{CM} \Rightarrow \boxed{a_{CM} = \frac{F/m}{1 + \frac{I_{CM}}{mR^2}}}$$

$$f_{as} = \frac{I_{CM}}{R^2} \frac{F/m}{1 + \frac{I_{CM}}{mR^2}} = \frac{F}{1 + \frac{mR^2}{I_{CM}}}$$

$$f_{as} \leq f_{as,max} = \mu_s N$$

$$\Rightarrow f_{as} = \frac{F}{1 + \frac{mR^2}{I_{CM}}} \leq \mu_s mg \Rightarrow$$

$$\Rightarrow \boxed{F \leq F_{max} = \mu_s mg \left(1 + \frac{mR^2}{I_{CM}}\right)}$$



$$\begin{cases} m\bar{g} + \bar{N} + \bar{f}_{as} = m\bar{a}_{CM} \\ \bar{r} \times \bar{f}_{as} + \bar{M} = \bar{I}_{CM}\bar{\alpha} \end{cases}$$

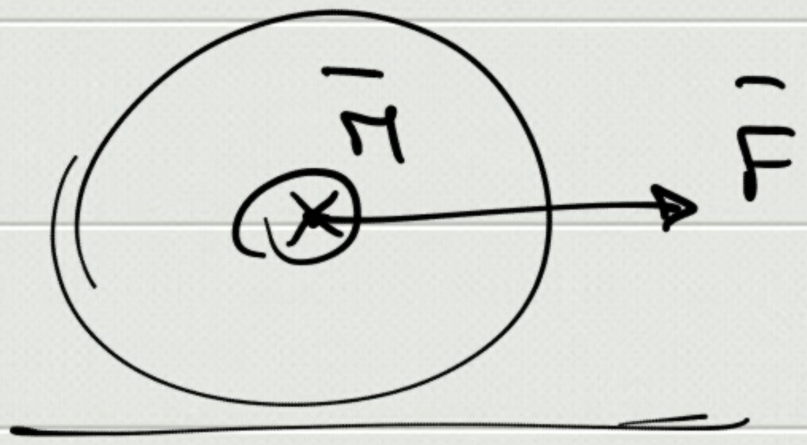
$$\Rightarrow \begin{cases} f_{as} = m a_{CM} \qquad\qquad N = mg \\ -R f_{as} + M = I_{CM}\alpha = I_{CM}\dfrac{a_{CM}}{R} \end{cases}$$

$$\Rightarrow M - mRa_{CM} = I_{CM}\frac{a_{CM}}{R}$$

$$\Rightarrow \boxed{a_{CM} = \frac{M/mR}{1 + \dfrac{I_{CM}}{mR^2}}}$$

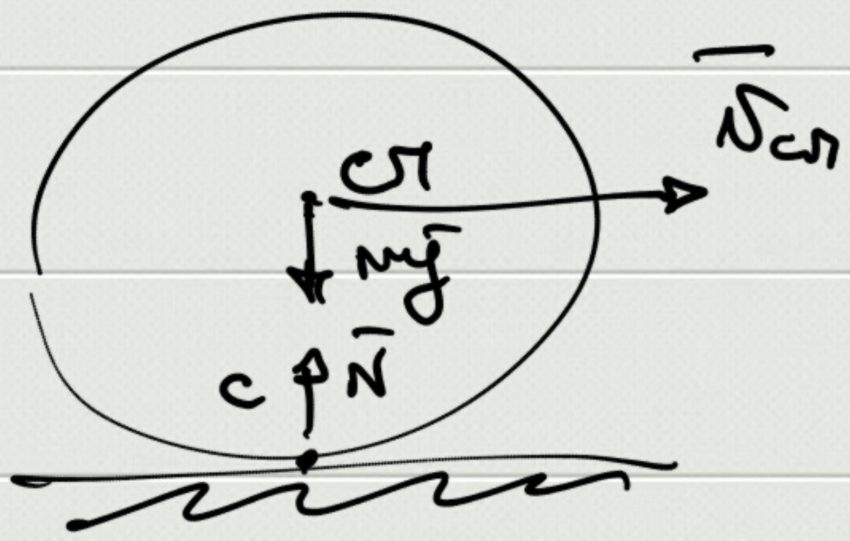
$$\Rightarrow \boxed{f_{as} = \frac{M/R}{1 + \dfrac{I_{CM}}{mR^2}}} \qquad \leq f_{as,max} = \mu_s mg$$

$$\Rightarrow \quad M \leq \mu_s m g R \left(1 + \frac{I_{CM}}{mR^2}\right)$$



$$a_{CM} = \frac{1}{m} \frac{M/R + F}{1 + \frac{I_{CM}}{mR^2}} \qquad f_{ps} = \frac{M/R - \frac{I_{CM}}{mR^2} F}{1 + \frac{I_{CM}}{mR^2}}$$

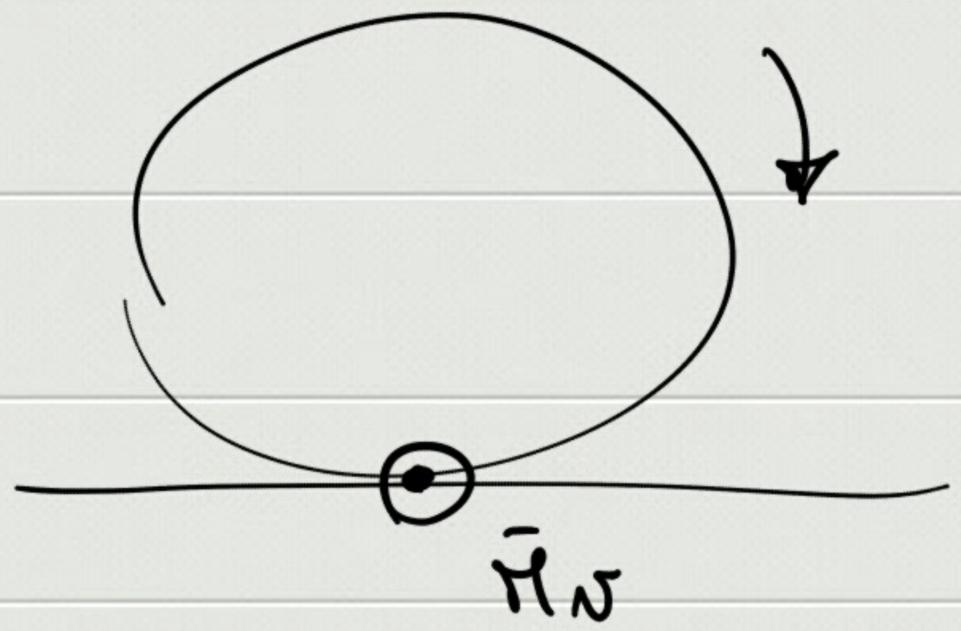
$$W_{TOT} = \Delta E_K$$



$$W = \int \bar{F}\, d\bar{s}$$

$$\Rightarrow \quad W_{f_{ps}} = \int \bar{f}_{ps}\, d\bar{s} = 0$$

Attrito volvente



$$\bar{M}_v = -h N \bar{v}_\omega$$

coeff. di attrito volvente

generalmente è trascurabile